

A uniform upper bound for Hamiltonian graphs of minimum degree three

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Abstract

For every integer $n \geq 19$, we construct a finite simple Hamiltonian graph of order n , size $2n - 5$, and minimum degree exactly three, with no cycle of length $n - 2$. Thus the eventual-order threshold $f(3)$ satisfies $f(3) \leq 19$, improving the uniform upper bound 31 in Zhan's construction. The family extends Zhan's known order-19 example by a path with a common neighbor. A fan reduction reduces the forbidden-cycle assertion to three base graphs, verified by exact enumeration of all 571 vertex-pair deletions. This note proves the upper bound; it supplies no lower-bound classification and leaves the original exact-value problem unresolved within its scope.

1 Statement and context

All graphs are finite, simple, and undirected. A cycle is simple and need not be induced; its length is its number of vertices. A graph is Hamiltonian if it has a cycle through all its vertices. Write $\delta(G)$ for its minimum degree. Following Zhan's Problem 2 [1], let $f(3)$ be the least positive integer p such that, for every integer $n \geq p$, there is a Hamiltonian graph of order n with minimum degree exactly three and no cycle of length $n - 2$.

Theorem 1. *For every integer $n \geq 19$, there exists a Hamiltonian graph G with $|V(G)| = n$, $|E(G)| = 2n - 5$, and $\delta(G) = 3$ that contains no cycle of length $n - 2$. Consequently, $f(3) \leq 19$.*

Zhan's Theorem 1 gives such graphs of minimum degree d for every $n \geq 15d - 14$, yielding $f(3) \leq 31$ [1]. The same paper displays an order-19, size-33 example in Figure 2. We credit that seed to Zhan. The contribution here is its explicit extension to every order at least 19, together with a reduction to three finite checks. An isolated example at order 19 alone does not establish this eventual-order bound.

The proof combines a structural reduction with exact finite verification. For adjacent work on prescribed degrees, cycle restrictions, and finite reductions in graph theory, see Zeng [2]; that reference concerns regular graphs of prescribed girth and is not a premise of our theorem. The comparison with prior Hamiltonian constructions is restricted to the inspected version of [1]; we make no exhaustive priority claim.

Zhan also reports an exhaustive minimality computation for the order-19 example [1]. We neither reproduce that classification nor use it in the proof. In particular, we do not claim the exact equality $f(3) = 19$.

2 The explicit family

Let H have vertex set $\{1, 2\} \cup \{5, 6, \dots, 19\}$ and the following 28 edges, where (u, v) denotes the unordered pair $\{u, v\}$:

$$\begin{aligned} E(H) = \{ & (1, 2), (1, 5), (1, 13), (1, 14), (1, 18), (1, 19), \\ & (2, 6), (5, 6), (6, 7), (6, 10), (7, 8), (7, 11), \\ & (8, 9), (8, 16), (9, 10), (9, 16), (10, 11), (11, 12), \\ & (11, 15), (12, 13), (12, 16), (13, 14), (14, 15), \\ & (15, 16), (16, 17), (17, 18), (17, 19), (18, 19)\}. \end{aligned} \quad (1)$$

For each integer $t \geq 2$, add distinct vertices $p_1, \dots, p_t$ outside H and define G_t by

$$\begin{aligned} V(G_t) &= V(H) \cup \{p_1, \dots, p_t\}, \\ E(G_t) &= E(H) \cup \{p_i p_{i+1} : 1 \leq i < t\} \\ &\quad \cup \{2p_1, 5p_t\} \cup \{11p_i : 1 \leq i \leq t\}. \end{aligned} \quad (2)$$

The graph G_2 is the known seed, with $p_1 = 3$ and $p_2 = 4$. The vertex and edge sets in this formula directly give

$$|V(G_t)| = 17 + t, \quad |E(G_t)| = 28 + (t - 1) + 2 + t = 2(17 + t) - 5.$$

The degrees of 2 and 5 in H are two, and that of 11 is four; every other core vertex has degree at least three. Hence 2 and 5 have degree three in G_t , vertex 11 has degree $t + 4$, and every added path vertex has degree three, including when $t = 2$. Thus $\delta(G_t) = 3$. Finally,

$$1, 2, p_1, \dots, p_t, 5, 6, 7, \dots, 19, 1 \quad (3)$$

is a Hamilton cycle: every consecutive pair is an edge in (1) or (2). It remains to exclude cycles that omit exactly two vertices.

3 A fan reduction

Lemma 2. *Let H be a finite simple graph with $h \geq 4$ vertices and three distinct specified vertices a, b, c . For $t \geq 2$, form F_t by adding a disjoint path $p_1 \cdots p_t$, the edges ap_1 and bp_t , and every edge cp_i . If F_2, F_3, F_4 have no cycle omitting exactly two vertices, then the same holds for every F_t , $t \geq 2$.*

Proof. It suffices to consider $t \geq 5$. Suppose that a cycle C in F_t omits exactly two vertices. Let r be the number of omitted path vertices. Then $0 \leq r \leq 2$, and C omits $2 - r$ core vertices. It meets the added path, since $t \geq 5$, and it meets the core outside c , since $h - 1 \geq 3$ core vertices differ from c .

Consider the subgraph consisting of the path vertices on C and the path edges used by C . Each component is a nonempty interval, possibly a singleton, with exactly two incident edges of C leading to the core. There are at most four such boundary edges: one through a , one through b , and two through c . Thus there are at most two intervals. An interval cannot have both its attachments at c : together with c it would already close the whole cycle, excluding the required core vertex outside c . For a singleton, two edges to c are impossible because the graph is simple. There are consequently three cases.

One interval attached at a and b . The interval contains both path endpoints, so it is the entire path and $r = 0$. Replace it by the whole two-vertex path of F_2 , retaining all used core vertices and edges. The resulting cycle omits the same two core vertices.

One interval attached at a and c , or at b and c . In the first case the interval is a prefix, with all r omitted path vertices in the suffix. If $r = 0$, replace it by the whole path of F_2 , using the a -edge at its first vertex and the c -edge at its last vertex. If $r = 1$, use only the first path vertex of F_2 . If $r = 2$, use only the first path vertex of F_3 . These choices omit respectively zero, one, or two path vertices and retain the original core subpath. The b, c case uses suffixes in the same way.

Two intervals. All four available boundary edges are used. Since neither interval has both attachments at c , one joins a to c and the other joins c to b . They are a prefix and a suffix, and the r omitted path vertices lie between them. Replace them by the first and last path vertices of F_{r+2} , respectively, omitting the r intermediate vertices. The attachments a, p_1, c and c, p_{r+2}, b , together with the retained a -to- b core subpath avoiding c , form a simple cycle. This also works when $r = 0$: the edge between the two path vertices is present but need not belong to the cycle.

In every case, the replacement retains the used core vertices, omits exactly r path vertices and $2 - r$ core vertices, and gives a simple cycle in one of F_2, F_3, F_4 . This contradicts the hypothesis. $\square$

4 The three base cases

Proposition 3. *For $t = 2, 3, 4$, the graph G_t has no cycle of length $|V(G_t)| - 2$.*

Proof by exact finite computation. An n -vertex graph, $n \geq 5$, has an $(n - 2)$ -cycle if and only if deleting some unordered pair of vertices leaves a graph with a Hamilton cycle. Thus it suffices to test every pair in each of G_2, G_3, G_4 .

Here is the exact counting recurrence used for those tests. For an induced graph K obtained by deleting a pair, choose its least-labelled vertex s . For $s, v \in A \subseteq V(K)$, let $D(A, v)$ count the simple paths from s to v with vertex set exactly A . Initialize $D(\{s\}, s) = 1$ and all other entries to zero. In increasing order of $|A|$, for every $w \in N_K(v) \setminus A$, perform

$$D(A \cup \{w\}, w) += D(A, v). \quad (4)$$

Every path has a unique predecessor obtained by removing its last vertex, so induction on $|A|$ proves the counting interpretation. The number of undirected Hamilton cycles is

$$\frac{1}{2} \sum_{v \in N_K(s)} D(V(K), v). \quad (5)$$

The two orientations of each cycle explain the factor $1/2$. If K has a vertex of degree less than two, its count is immediately zero.

Applying this recurrence with exact integers to the edge lists specified by (1)–(2) gives Table 1. All pair deletions are included, and every count is zero. The supplementary program `verify_bases.py` constructs these edge lists directly and implements (4)–(5); it uses only the Python standard library. Its output `verification.json` records all $171 + 190 + 210 = 571$ pair counts, together with graph and control checks. The finite computation completes without sampling, floating-point arithmetic, a search cutoff, or a timeout-based conclusion. $\square$

Proof of Theorem 1. Use the core (1) and $(a, b, c) = (2, 5, 11)$ in Lemma 2. Proposition 3 supplies its three base cases, so G_t has no cycle omitting exactly two vertices for every $t \geq 2$. The order, size, minimum degree, and Hamiltonicity were verified in Section 2. Taking $t = n - 17$ proves every assertion for every $n \geq 19$. $\square$

Table 1: Complete base-case verification. The final column sums the Hamilton-cycle counts over all vertex-pair deletions.

t	Order	Size	Minimum degree	Pairs tested	Cycle count
2	19	33	3	171	0
3	20	35	3	190	0
4	21	37	3	210	0

5 Verification and scope

The base cases were also checked by a separate simple-path depth-first enumeration. A further implementation, written in a separate mathematical review pass, enumerated spanning edge sets of degree two using edge inclusion and exclusion with degree propagation, and tested connectivity at completion. A connected spanning 2-regular graph is a Hamilton cycle; the connectivity test is essential. Across the pair deletions, that enumeration found respectively 236, 247, and 256 spanning 2-factors, all disconnected, and zero Hamilton cycles. These independent implementations agree with the subset recurrence.

Positive controls for the recurrence give $(m - 1)!/2$ Hamilton cycles in K_m for $3 \leq m \leq 7$, and one in C_5, C_7, C_9 . Negative controls include two triangles sharing one vertex and two disjoint triangles. The supplementary README gives the command for repeating the finite verification. No enumeration of all graphs of any order is claimed.

The result is a uniform upper bound for minimum degree exactly three. It does not assert regularity, edge-minimality, or the absence of cycles of lengths other than $n - 2$. In particular, there is no order-18 nonexistence proof here. Determining the exact threshold is outside the proved scope, even though the primary source reports a lower-bound classification. The theorem itself requires only the explicit construction, the fan lemma, and the exact base computations given above.

Computational and AI assistance. OpenAI Codex assisted with the construction search, the mathematical argument, verification programs, source comparison, and manuscript preparation. The separate mathematical review pass used a fresh model context and a different finite-verification algorithm. This is an additional computational and mathematical check, not human peer review or a claim of statistical independence. The proof is computer-assisted only in its explicitly delimited three finite base cases; the extension to all orders is the deductive argument of Lemma 2.

References

- [1] Xingzhi Zhan, *Hamiltonian graphs with prescribed minimum degree and no near-spanning cycles*, arXiv:2609.05468v1 (2026). <https://arxiv.org/abs/2609.05468v1>.
- [2] Zijian Zeng, *Proof-Carrying Reductions for 4-Regular Girth-Nine Graphs of Excess Four*, SSRN (2026). <https://doi.org/10.2139/ssrn.7374858>.